Group 0 – the Noble Gases

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 8 · 16 marks · about 25 minutes

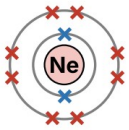
Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

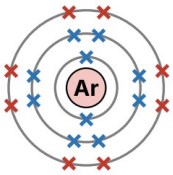
- The group 0 elements are called the **noble gases**: helium, neon, argon, krypton, xenon and radon.
- They are **unreactive** (they do not easily form molecules) because they have a **full outer shell** of electrons – a stable arrangement.
- Helium has **2** electrons in its outer shell. The others have **8**.
- They exist as **single atoms** (they are monatomic).
- Going down the group the **relative atomic mass increases**, so the **boiling points increase**.



helium 2



neon 2,8



argon 2,8,8

Boiling points of the noble gases

Noble gas	Relative atomic mass	Boiling point in °C
helium	4	−269
neon	20	−246
argon	40	−186
krypton	84	−153
xenon	131	−108
radon	222	?

Section A: Quick recall (6 marks – 1 mark each)

1 What name is given to the group 0 elements?

2 Why are the noble gases unreactive?

3 How many electrons are in the outer shell of a helium atom?

4 How many electrons are in the outer shell of a neon atom?

5 Do noble gases exist as single atoms or as molecules?

6 What happens to the boiling point going down group 0?

Section B: Exam questions

01 Argon has the electronic structure 2,8,8. Explain why argon is unreactive.

[2 marks]

02 Use the table of boiling points.

02.1 Describe the trend in boiling point going down group 0.

[1 mark]

02.2 Predict the boiling point of radon.

[1 mark]

03 The noble gases were not included in Mendeleev's original periodic table. Suggest why.

[2 marks]

04 Draw a diagram to show the electronic structure of a neon atom (atomic number 10).

[2 marks]

05 Helium is used to fill party balloons. Hydrogen is less dense than helium but is not used. Suggest why helium is used instead of hydrogen.

[2 marks]

Mark Scheme – Group 0 – the Noble Gases

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** the noble gases **2** they have a full outer shell (stable arrangement of electrons) **3** 2 **4** 8 **5** single atoms **6** it increases	1 mark each	6
01	it has a full outer shell (of 8 electrons) (1) so it does not need to lose, gain or share electrons / it is stable (1)		2
02.1	boiling point increases (1)		1
02.2	any value between –50 °C and –80 °C (actual value –62 °C) (1)		1
03	they had not been discovered yet (1) because they are unreactive / do not form compounds, so they were hard to detect (1)		2
04	two shells drawn around a nucleus (1) 2 electrons in the first shell and 8 in the second shell (1)		2
05	helium is unreactive / will not burn (1) hydrogen is flammable / could explode (1)		2

Total: 16 marks